

# Storage Devices -- Guided Notes

Unit tp-2.3 | CompTIA Network+ N10-009 Obj. FC0-U71 2.3 | N10-008 FC0-U71 2.3

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

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Complete each question using the lesson reading. Write in the blank or answer in the space provided.

## Question 1

A HDD (Hard Disk Drive) stores data on \_\_\_\_\_ platters that spin at speeds like 5,400 or 7,200 \_\_\_\_\_. A read/write \_\_\_\_\_ moves across the platters to access data. Because of these moving parts, HDDs are more susceptible to \_\_\_\_\_ damage from dropping.

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## Question 2

An SSD (Solid State Drive) stores data using \_\_\_\_\_ (NAND flash) chips with no moving parts. Compared to an HDD, an SSD is \_\_\_\_\_ (faster/slower), \_\_\_\_\_ (more/less) resistant to physical shock, and \_\_\_\_\_ (more/less) expensive per gigabyte.

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## Question 3

An M.2 NVMe SSD connects directly to the \_\_\_\_\_ bus via a slot on the motherboard, achieving speeds of \_\_\_\_\_ to \_\_\_\_\_ GB/s. A SATA SSD is limited to approximately \_\_\_\_\_ MB/s because SATA was designed for spinning hard drives.

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## Question 4

SATA (Serial AT Attachment) is the interface used by most \_\_\_\_\_ and 2.5-inch \_\_\_\_\_ drives. SATA III has a maximum speed of \_\_\_\_\_ Gbps (approximately 600 MB/s). The SATA cable carries \_\_\_\_\_ only; a separate power cable connects from the PSU.

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## Question 5

An optical drive reads and writes data to \_\_\_\_\_ using a laser. Common formats include CD (\_\_\_\_\_ MB), DVD (\_\_\_\_\_ GB), and Blu-ray (\_\_\_\_\_ GB). Optical drives are becoming less common in modern systems because \_\_\_\_\_ and digital distribution have replaced physical media.

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## Question 6

RAID (Redundant Array of Independent Disks) combines multiple drives for performance or redundancy. RAID 0 (striping) \_\_\_\_\_ performance but provides \_\_\_\_\_ redundancy -- if one drive fails, all data is lost. RAID 1 (mirroring) duplicates data across \_\_\_\_\_ drives for redundancy at the cost of \_\_\_\_\_ the usable storage.

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#### Question 7

NAS (Network Attached Storage) is a dedicated storage device connected to a network, allowing multiple \_\_\_\_\_ to access shared files. It differs from DAS (Direct Attached Storage) in that DAS connects \_\_\_\_\_ to a single computer, while NAS is accessible to anyone on the \_\_\_\_\_.

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#### Question 8 -- Real World Application

REAL WORLD: A school is upgrading 40 student laptops that currently use 256 GB HDDs. The budget allows for either SATA SSDs or NVMe SSDs as replacements. The primary use case is running Windows, a web browser, and Office applications. Which storage upgrade would you recommend and why? Is NVMe worth the additional cost for this scenario?

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*Download the answer key from the class server:*

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wget http://[SERVER]/resources/network-engineer/tp-2-3-storage-devices/tp-2-3-storage-devices-answer-key.pdf
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